الجامعة الاسلامية العالمية

International Islamic University, Islamabad

CAR RENTAL MANAGEMENT SYSTEM – ERD REPORT

(Database Design Phase – Part 1)

Submitted by:

Farhan Ullah Khan (4764-FOC/BSSE/F23)

Mohammad Sami (4835-FOC/BSSE/F23)

Taha malik (4814-FOC/BSSE/F23)

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Instructor: Mr. Shakir Rasheed

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Department of software engineering

Faculty of Computing and Information Technology

INTERNATIONAL ISLAMIC UNIVERSITY, ISLAMABAD

Car Rental Management System

Entity Relationship Diagram (ERD) Documentation

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1. PROJECT OVERVIEW

1.1 Introduction

The Car Rental Management System is a comprehensive database solution designed to manage the core operations of a modern car rental business. This system facilitates efficient management of vehicle inventory, customer information, rental reservations, payment processing, vehicle maintenance, and employee records.

1.2 Objectives

The primary objectives of this system are:

- **Customer Management**: Maintain comprehensive customer profiles including personal information, licensing details, and rental history
- **Vehicle Fleet Management**: Track vehicle availability, specifications, categories, and operational status
- **Reservation Processing**: Handle booking requests, schedule pickups and returns, and manage reservation lifecycle
- **Payment Processing**: Process rental payments, track transaction history, and manage additional charges
- **Maintenance Tracking**: Monitor vehicle maintenance schedules, service history, and associated costs
- **Employee Management**: Maintain employee records and track their involvement in reservations and maintenance activities

1.3 Scope

This system encompasses end-to-end rental operations from initial customer registration through vehicle reservation, rental period management, payment collection, and ongoing vehicle maintenance. It supports multiple locations, various vehicle categories, and different payment methods while maintaining data integrity and business rule enforcement.

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2. ENTITIES AND ATTRIBUTES

2.1 CUSTOMER Entity

Description: Stores information about individuals or organizations that rent vehicles from the company.

Attributes:

- **customer_id** (PK, INT): Unique identifier for each customer
- **first_name** (VARCHAR(50)): Customer's first name
- **last_name** (VARCHAR(50)): Customer's last name
- **email** (VARCHAR(100), UNIQUE): Customer's email address for communication
- **phone** (VARCHAR(15)): Contact phone number
- **license_number** (VARCHAR(20), UNIQUE): Driver's license number
- **license_expiry** (DATE): Expiration date of driver's license
- **address** (VARCHAR(200)): Street address
- **city** (VARCHAR(50)): City of residence
- **state** (VARCHAR(50)): State/province of residence
- **zip_code** (VARCHAR(10)): Postal code
- **registration_date** (DATE): Date when customer registered with the system
- **customer_type** (VARCHAR(20)): Type of customer (Individual, Corporate, VIP)

Primary Key: customer_id

2.2 VEHICLE Entity

Description: Contains detailed information about each vehicle in the rental fleet.

Attributes:

- **vehicle_id** (PK, INT): Unique identifier for each vehicle
- **registration_number** (VARCHAR(20), UNIQUE): Vehicle registration/license plate number
- **category_id** (FK, INT): Reference to vehicle category
- **make** (VARCHAR(50)): Vehicle manufacturer (e.g., Toyota, Ford)
- **model** (VARCHAR(50)): Vehicle model name
- **year** (INT): Manufacturing year
- **color** (VARCHAR(30)): Vehicle color
- **mileage** (INT): Current odometer reading

- **status** (VARCHAR(20)): Current status (Available, Rented, Maintenance, Retired)
- **last_service_date** (DATE): Date of most recent maintenance
- **daily_rate** (DECIMAL(10,2)): Rental rate per day
- **fuel_type** (VARCHAR(20)): Type of fuel (Gasoline, Diesel, Electric, Hybrid)
- **transmission_type** (VARCHAR(20)): Transmission type (Automatic, Manual)

Primary Key: vehicle_id
Foreign Key: category_id references VEHICLE_CATEGORY(category_id)

2.3 VEHICLE_CATEGORY Entity

Description: Defines different categories of vehicles available for rent, each with specific characteristics and pricing.

Attributes:

- **category_id** (PK, INT): Unique identifier for each vehicle category
- **category_name** (VARCHAR(50), UNIQUE): Name of category (Economy, Compact, SUV, Luxury, Van)
- **description** (TEXT): Detailed description of category features
- **passenger_capacity** (INT): Maximum number of passengers
- **base_rate** (DECIMAL(10,2)): Base daily rental rate for this category
- **insurance_rate** (DECIMAL(10,2)): Daily insurance cost for this category

Primary Key: category_id

2.4 RESERVATION Entity

Description: Records all rental reservations made by customers, including booking details and rental period information.

Attributes:

- **reservation_id** (PK, INT): Unique identifier for each reservation
- **customer_id** (FK, INT): Reference to customer making reservation
- **vehicle_id** (FK, INT): Reference to reserved vehicle

- **employee_id** (FK, INT): Reference to employee processing reservation
- **reservation_date** (DATE): Date when reservation was created
- **pickup_date** (DATE): Scheduled vehicle pickup date
- **return_date** (DATE): Scheduled vehicle return date
- **actual_return_date** (DATE): Actual date vehicle was returned (may differ from scheduled)
- **pickup_location** (VARCHAR(100)): Location where vehicle will be picked up
- **return_location** (VARCHAR(100)): Location where vehicle will be returned
- **total_amount** (DECIMAL(10,2)): Total calculated rental cost
- **status** (VARCHAR(20)): Reservation status (Pending, Confirmed, Active, Completed, Cancelled)
- **special_requests** (TEXT): Any special requirements or notes

Primary Key: reservation_id
Foreign Keys:

- customer_id references CUSTOMER(customer_id)
- vehicle_id references VEHICLE(vehicle_id)
- employee_id references EMPLOYEE(employee_id)

2.5 PAYMENT Entity

Description: Tracks all payment transactions related to rental reservations.

Attributes:

- **payment_id** (PK, INT): Unique identifier for each payment
- **reservation_id** (FK, INT): Reference to associated reservation
- **customer_id** (FK, INT): Reference to customer making payment
- **payment_date** (DATE): Date payment was processed
- **amount** (DECIMAL(10,2)): Payment amount
- **payment_method** (VARCHAR(30)): Method of payment (Credit Card, Debit Card, Cash, Bank Transfer)
- **transaction_id** (VARCHAR(100), UNIQUE): Unique transaction reference number
- **payment_status** (VARCHAR(20)): Status of payment (Pending, Completed, Failed, Refunded)

- **late_fee** (DECIMAL(10,2)): Additional charges for late return
- **damage_charge** (DECIMAL(10,2)): Charges for vehicle damage

Primary Key: payment_id
Foreign Keys:

- reservation_id references RESERVATION(reservation_id)
- customer_id references CUSTOMER(customer_id)

2.6 MAINTENANCE Entity

Description: Records all maintenance and service activities performed on vehicles.

Attributes:

- **maintenance_id** (PK, INT): Unique identifier for each maintenance record
- **vehicle_id** (FK, INT): Reference to vehicle being serviced
- **employee_id** (FK, INT): Reference to employee responsible for maintenance
- **maintenance_date** (DATE): Date maintenance was performed
- **maintenance_type** (VARCHAR(50)): Type of maintenance (Routine Service, Repair, Inspection, Cleaning)
- **description** (TEXT): Detailed description of work performed
- **cost** (DECIMAL(10,2)): Cost of maintenance
- **status** (VARCHAR(20)): Status of maintenance (Scheduled, In Progress, Completed)
- **next_service_date** (DATE): Scheduled date for next service

Primary Key: maintenance_id
Foreign Keys:

- vehicle_id references VEHICLE(vehicle_id)
- employee_id references EMPLOYEE(employee_id)

2.7 EMPLOYEE Entity

Description: Stores information about employees who manage reservations and vehicle maintenance.

Attributes:

- **employee_id** (PK, INT): Unique identifier for each employee
- **first_name** (VARCHAR(50)): Employee's first name
- **last_name** (VARCHAR(50)): Employee's last name
- **email** (VARCHAR(100), UNIQUE): Employee's company email
- **phone** (VARCHAR(15)): Contact phone number
- **position** (VARCHAR(50)): Job title/position
- **hire_date** (DATE): Date employee was hired
- **salary** (DECIMAL(10,2)): Employee's salary
- **department** (VARCHAR(50)): Department (Rental Operations, Maintenance, Administration)
- **status** (VARCHAR(20)): Employment status (Active, On Leave, Terminated)

Primary Key: employee_id

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3. RELATIONSHIPS

3.1 CUSTOMER to RESERVATION (1:M)

- **Relationship Type**: One-to-Many
- **Description**: A customer can make multiple reservations over time, but each reservation belongs to exactly one customer
- **Cardinality**: 1:M
- **Participation**:
  - Customer: Partial (a customer may exist without reservations)
  - Reservation: Total (every reservation must be associated with a customer)

3.2 CUSTOMER to PAYMENT (1:M)

- **Relationship Type**: One-to-Many
- **Description**: A customer can make multiple payments for different reservations, but each payment is made by exactly one customer
- **Cardinality**: 1:M
- **Participation**:

- Customer: Partial (a customer may exist without payments)
- Payment: Total (every payment must be associated with a customer)

3.3 VEHICLE to RESERVATION (1:M)

- **Relationship Type**: One-to-Many
- **Description**: A vehicle can be reserved multiple times (at different periods), but each reservation involves exactly one vehicle
- **Cardinality**: 1:M
- **Participation**:
  - Vehicle: Partial (a vehicle may exist without being reserved)
  - Reservation: Total (every reservation must include a vehicle)

3.4 VEHICLE_CATEGORY to VEHICLE (1:M)

- **Relationship Type**: One-to-Many
- **Description**: A vehicle category can include multiple vehicles, but each vehicle belongs to exactly one category
- **Cardinality**: 1:M
- **Participation**:
  - Vehicle Category: Partial (a category may exist without vehicles)
  - Vehicle: Total (every vehicle must belong to a category)

3.5 VEHICLE to MAINTENANCE (1:M)

- **Relationship Type**: One-to-Many
- **Description**: A vehicle can undergo multiple maintenance activities over time, but each maintenance record pertains to exactly one vehicle
- **Cardinality**: 1:M
- **Participation**:
  - Vehicle: Partial (a vehicle may exist without maintenance records)

- Maintenance: Total (every maintenance record must be associated with a vehicle)

3.6 RESERVATION to PAYMENT (1:1)

- **Relationship Type**: One-to-One
- **Description**: Each reservation generates exactly one primary payment transaction, and each payment is associated with exactly one reservation
- **Cardinality**: 1:1
- **Participation**:
  - Reservation: Total (every reservation must have a payment)
  - Payment: Total (every payment must be linked to a reservation)

3.7 EMPLOYEE to RESERVATION (1:M)

- **Relationship Type**: One-to-Many
- **Description**: An employee can process multiple reservations, but each reservation is processed by exactly one employee
- **Cardinality**: 1:M
- **Participation**:
  - Employee: Partial (an employee may exist without processing reservations)
  - Reservation: Total (every reservation must be processed by an employee)

3.8 EMPLOYEE to MAINTENANCE (1:M)

- **Relationship Type**: One-to-Many
- **Description**: An employee can perform multiple maintenance activities, but each maintenance record is performed by exactly one employee
- **Cardinality**: 1:M
- **Participation**:
  - Employee: Partial (an employee may exist without maintenance records)
  - Maintenance: Total (every maintenance record must be performed by an employee)

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4. ASSUMPTIONS

The following assumptions have been made during the design of this Car Rental Management System:

1. **Customer Registration**: All customers must complete registration and provide valid driver's license information before making reservations.
2. **Single Vehicle per Reservation**: Each reservation is for exactly one vehicle. Customers wanting multiple vehicles must create separate reservations.
3. **Payment Processing**: A primary payment is recorded for each reservation at the time of booking or vehicle pickup. Additional charges (late fees, damage charges) are added to the same payment record.
4. **Vehicle Availability**: A vehicle can only be assigned to one active reservation at a time. The system checks availability based on reservation dates before confirming bookings.
5. **Employee Roles**: All employees can process reservations and perform maintenance. Specific role-based access control would be implemented at the application level.
6. **Maintenance Scheduling**: Vehicles undergo regular maintenance based on mileage and time intervals. Vehicles in maintenance status are not available for rental.
7. **Location Management**: Pickup and return locations are stored as text attributes. A separate LOCATION entity could be added for more complex multi-branch operations.
8. **License Validation**: The system assumes that driver's license verification is performed during registration, and licenses must be valid throughout the rental period.
9. **Pricing Calculation**: Total rental amount is calculated based on the number of days, vehicle category base rate, and any additional services. The calculation logic is handled by the application layer.
10. **Data Retention**: Historical records (completed reservations, past maintenance) are retained indefinitely for business intelligence and audit purposes.
11. **Unique Constraints**: Email addresses and license numbers are unique across the system to prevent duplicate customer registrations.
12. **Date Validation**: Return dates must be after pickup dates, and actual return dates are recorded when vehicles are returned.

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5. CONSTRAINTS

5.1 Primary Key Constraints

- **CUSTOMER**: customer_id (auto-increment)

- **VEHICLE**: vehicle_id (auto-increment)
- **VEHICLE_CATEGORY**: category_id (auto-increment)
- **RESERVATION**: reservation_id (auto-increment)
- **PAYMENT**: payment_id (auto-increment)
- **MAINTENANCE**: maintenance_id (auto-increment)
- **EMPLOYEE**: employee_id (auto-increment)

5.2 Foreign Key Constraints

- **VEHICLE.category_id** → VEHICLE_CATEGORY.category_id (ON DELETE RESTRICT, ON UPDATE CASCADE)
- **RESERVATION.customer_id** → CUSTOMER.customer_id (ON DELETE RESTRICT, ON UPDATE CASCADE)
- **RESERVATION.vehicle_id** → VEHICLE.vehicle_id (ON DELETE RESTRICT, ON UPDATE CASCADE)
- **RESERVATION.employee_id** → EMPLOYEE.employee_id (ON DELETE RESTRICT, ON UPDATE CASCADE)
- **PAYMENT.reservation_id** → RESERVATION.reservation_id (ON DELETE RESTRICT, ON UPDATE CASCADE)
- **PAYMENT.customer_id** → CUSTOMER.customer_id (ON DELETE RESTRICT, ON UPDATE CASCADE)
- **MAINTENANCE.vehicle_id** → VEHICLE.vehicle_id (ON DELETE RESTRICT, ON UPDATE CASCADE)
- **MAINTENANCE.employee_id** → EMPLOYEE.employee_id (ON DELETE RESTRICT, ON UPDATE CASCADE)

5.3 Unique Constraints

- **CUSTOMER**: email, license_number
- **VEHICLE**: registration_number
- **VEHICLE_CATEGORY**: category_name
- **PAYMENT**: transaction_id
- **EMPLOYEE**: email

5.4 Check Constraints

- **CUSTOMER.license_expiry** > CURRENT_DATE (license must be valid)
- **VEHICLE.year** BETWEEN 2000 AND CURRENT_YEAR + 1
- **VEHICLE.mileage** >= 0
- **VEHICLE.status** IN ('Available', 'Rented', 'Maintenance', 'Retired')
- **RESERVATION.return_date** > RESERVATION.pickup_date
- **RESERVATION.status** IN ('Pending', 'Confirmed', 'Active', 'Completed', 'Cancelled')
- **PAYMENT.amount** > 0
- **PAYMENT.payment_status** IN ('Pending', 'Completed', 'Failed', 'Refunded')
- **MAINTENANCE.cost** >= 0
- **EMPLOYEE.salary** > 0

5.5 Not Null Constraints

All primary keys and foreign keys must be NOT NULL. Additionally:

- Customer: first_name, last_name, email, phone, license_number
- Vehicle: registration_number, category_id, make, model, status
- Reservation: customer_id, vehicle_id, reservation_date, pickup_date, return_date, status
- Payment: reservation_id, customer_id, payment_date, amount, payment_method
- Employee: first_name, last_name, email, position

5.6 Domain Constraints

- Email addresses must follow standard email format
- Phone numbers must contain only digits and standard separators
- Dates must be valid calendar dates
- Monetary values (DECIMAL) must have exactly 2 decimal places
- Status fields must contain only predefined values

5.7 Business Rule Constraints

- A vehicle cannot have overlapping active reservations
- Customers must be at least 21 years old (enforced by application)

- Payment amount must match or exceed reservation total_amount
- Vehicles under maintenance cannot be reserved
- Actual return date cannot be before pickup date

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6. CONCLUSION

6.1 System Effectiveness

This Entity Relationship Diagram provides a robust foundation for a comprehensive Car Rental Management System that effectively supports real-world business operations. The design addresses all critical aspects of the car rental business lifecycle:

Reservation Management: The ERD facilitates seamless booking processes by connecting customers with available vehicles through a well-structured reservation system. The relationship between CUSTOMER, VEHICLE, and RESERVATION entities allows for efficient tracking of current and historical rentals, enabling the business to manage vehicle availability in real-time and prevent double-bookings.

Vehicle Fleet Management: Through the VEHICLE and VEHICLE_CATEGORY entities, the system maintains detailed records of the entire fleet, including specifications, current status, and categorization. This structure supports dynamic pricing based on vehicle categories and enables quick identification of available vehicles meeting specific customer requirements.

Payment Processing: The PAYMENT entity's one-to-one relationship with RESERVATION ensures accurate financial tracking for each rental transaction. The inclusion of fields for late fees and damage charges allows the system to handle common scenarios where additional costs are incurred, providing complete financial accountability.

Maintenance Operations: The MAINTENANCE entity ensures vehicles remain in optimal condition by tracking service history, scheduling future maintenance, and temporarily removing vehicles from the available fleet when necessary. This proactive approach minimizes breakdowns and enhances customer satisfaction.

Operational Control: Employee involvement in both reservations and maintenance activities creates an audit trail and accountability system. The EMPLOYEE entity enables the business to track productivity, manage workload distribution, and maintain service quality standards.

6.2 Scalability and Extensibility

The modular design allows for future enhancements such as:

- Addition of insurance packages and add-on services

- Implementation of loyalty programs and customer discounts
- Integration of location management for multi-branch operations
- Expansion to include damage assessment and claims processing
- Addition of customer reviews and feedback mechanisms

6.3 Data Integrity

The comprehensive constraint system ensures data accuracy and consistency across all operations. Foreign key relationships maintain referential integrity, while check constraints enforce business rules at the database level. Unique constraints prevent data duplication, and not-null constraints ensure critical information is always captured.

6.4 Practical Application

This ERD is production-ready and can be directly translated into a relational database implementation using any major RDBMS (MySQL, PostgreSQL, Oracle, SQL Server). The design follows normalization principles to minimize redundancy while maintaining query performance through logical relationship structures.

6.5 Final Assessment

The proposed Car Rental Management System ERD successfully models a complete business ecosystem that handles customer relationships, inventory management, financial transactions, and operational maintenance. It provides the data architecture necessary for developing a full-featured application that can compete in the modern car rental industry while maintaining flexibility for future business needs and technological advancements.

ER Diagram - Car Rental Management System